Ax=b
??
A
$n\times$
n
??
A
Ax=b

$$a_{11}a_{12}\cdots a_{1n}a_{21}a_{22}\cdots a_{2n}\quad a_{n1}a_{n2}\cdots a_{nn}x_1x_2x_n = b_1b_2b_n \tag{1}$$

x
A

$$a_{11}a_{12}\cdots a_{1n}0a_{22}\cdots a_{2n}\quad 00\cdots a_{mn}x_1x_2x_n = b_1b_2b_n \tag{2}$$

A

$$\mathbf{A} = 10\cdots 0a_{21}a_{11}1\cdots 0\quad a_{n1}a_{11}a_{n2}a_{22}\cdots 1\mathbf{U} \tag{3}$$

U
??
Ax=b
LUx=b
y=Ux
Ly=b
y
Ux=y
x
N
NB
$P\times$
Q
The size of the matrix Ns**:**
N
N
N
N
$N\times$
N
80%
$N\times$
$N\times$
$8=$
80%
The size of the matrix block(NBs**):**
NB
NB
$NB\times$
NB
NB
NB
NB
NB
Two-dimensional processor grid ($P\times$
Q**):**
$P\times$
Q
P
Q
Q
P
$P\times$
Q
$P\times$
$Q=$
$P\times$
$Q=$
$\overline{P} \leq$
Q
P
Q
ConfigurationPerformance
??
??
ConfigurationPerformance
??
Compile OptionsFunctional Description
LU
NameModelQuantity
??
The overall

??
ItemNameVersion
N
NB
$P\times$
Q
N=70000
NB=384$P\times$
Q=1$\times$
1
??
$Efficiency$=$\frac{PeakofHPL}{TheoreticalPeak}$=$\frac{869.00Gflops}{947.2Gflops}$≈91.74%

(6)
??
N=36864
NB=1024
$P\times$
Q=1$\times$
1
$Efficiency$=$\frac{PeakofHPL}{TheoreticalPeak}$=$\frac{632.1Gflops}{689Gflops}$≈91.74%

(7)
N
NB
$P\times$
Q
CPU
ef-
fi-
ciency
and
GPU
ef-
fi-
ciency
all
reached
about
91.74%